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Development in Bhutan: Happiness, Hydropolitics, and India

Background:

Between the looming powers of China and India, high up in the Eastern Himalayas, lies a small land-locked country with a population of only 800,000 people. For much of its history, this country was closed off to visitors, creating a pure cultural oasis and shrouding it in mysticism and mystery. The traditional name for this country is Druk Yul, meaning 'Land of the Thunder Dragon,' named after the wild thunderstorms from the peaks of the Himalayas that often strike the valley floors.

It takes very little effort to find an article describing Bhutan in a manner such as this imitation passage. While the imagery conjured by such a passage may be beautiful, a trend of these descriptions reveals a bad habit of sensationalization and exoticization practiced by most Western publications. Predictably, most journalistic treatment of Bhutan indulges in this exoticization, including non Western publications. An example of this can be found in A.C. Sinha's *Himalayan Kingdom Bhutan, Tradition, Transition and Transformation:*

> "Bhutan, the unspoiled, exotic and *Shangri-la* in the hidden Himalaya, is the only Buddhist Kingdom in the world. Known to its people as the *Dragon* Kingdom (*Brug or Grugyul),* she has stoon on guard as a trustee of an other-worldly tradition of *Lamaist* Buddhism. She possesses unparallelled and breath-taking scenic beauty and claims to be the only South Asian country without a population problem." (Sinha 2004)

It is through this lens of romanticism that the dominant narrative of Bhutan is crafted: a small, pristine, and culturally unique oasis juxtaposed by its larger neighboring countries. Under this narrative, Bhutan's developmental, technological and environmental advancements are framed as contrasting to the country's size and stature. The country has frequented national headlines in recent years for pledging to remain carbon neutral or negative, for their COVID vaccine roll-out "success story," and for drafting new blockchain policies intended to create a national digital currency -- all of which are achievements touted by major news outlets as being remarkably made *despite* Bhutan being one of the smallest and "least developed" countries in the world. (Tutton et al. 2018)

Bhutan's measure of development has also landed the country in headlines. Gross National Happiness (GNH), sometimes referred to as Gross Domestic Happiness (GDH) was determined as the guiding philosophy for Bhutan's government in 2008 when the Constitution of Bhutan was enacted. This document marked the transition from an absolute monarchy to a constitutional monarchy, making the post-2008 period a distinct separation from the rest of Bhutanese history. GNH was created in order to promote the values of Bhutanese culture drawn from citizen surveys and conveyed through four pillars and nine "domains of happiness." The nine domains of GNH are listed as physiological well-being, health, time use, education, cultural diversity and resilience, good governance, community vitality, ecological diversity and resilience, and living standards. Though the list of happiness domains may seem long to those of us with a more one-dimensional, Western social definition of happiness, the multifaceted approach is purposeful. Bhutan's policy makers consider happiness itself to be multifaceted, to be measured by more than just subjective well-being, and to be greater than individual happiness. Though happiness can be experienced personally, GNH is built on collective happiness, and is meant to orient the nation as a whole toward happiness.

The creation of GNH is attributed to the Fourth Druk Gyalpo (king) of Bhutan, Jigme Singye Wangchuck, who is famously quoted as declaring "Gross National Happiness is more important than Gross Domestic Product." At the time, Jigme Singye[1] was only sixteen years old and had just been crowned following his father's untimely death. He developed Gross National

[1] There is no such thing as surnames or family names in Bhutan, except for in the royal family. (Dorji 1976)

Happiness and continued his father's work of modernizing and democratizing Bhutan. Jigme Singye promoted the drafting of a constitution years before its creation. When questioned as to why it took so long for a constitution to be made, Jigme Singye responded: "I have been waiting for literacy rates to go up." In the years leading to 2008, Jigme Singye claims to have focused on improving primary education until literacy rates exceeded 60%. (Matsuzawa 2019)

The use of GNH as a measure of development makes Bhutan unique in that it is a rejection of GDP and other growth-focused development. Its creation reflects the ideologies held by the state, and that Bhutan's development leaders want to be seen following a specific value system. Being a Buddhist country, the Wangchuck monarchy's development policy is influenced by Mahayana Buddhism, which embraces the relationship between three interconnected aspects of human existence: human beings, society, and the natural environment.

> "Sustainable development has a special meaning and appeal in Bhutanese society, since it supports harmonious coexistence with the natural system, which is consistent with the common Buddhist beliefs. This relationship is manifested in peoples' beliefs that high mountains and deep ravines, ancient trees and rocks are the abode of spirits, gods and demons. Disturbing these elements would enrage these spirits and bring ill luck, sickness and death to families. By contrast, appeasing these spirits would be rewarded with luck, peace and prosperity.
> In line with these views, Bhutan's development strategy is guided by the philosophy of 'gross national happiness'. This concept expresses a preference for happiness over accumulation of material wealth and the development path therefore rests on the four so-called pillars of development: sustainable and equitable economic development; ecological preservation; cultural preservation; good governance." (Brooks 2010, pg. 54-55)

Based on these values, Brooks posits that Bhutan's development philosophy follows Buddhism's "middle way" or "middle path" in its development strategy. It can be inferred that the Bhutanese development strategy values moderate consumption for the betterment of quality of life over economic growth. There is a Bhutanese proverb that says: ‘it is better to have milk and cheese many times than beef once.” This sentiment carries over into policy, and Bhutan’s strategy is to ignore short-term gains for long-term benefits.

Bhutan’s commitment to sustainability through national policy exemplifies how Buddhist beliefs have influenced developmental philosophy. At a state level, it’s very clear that Bhutan values sustainability. These beliefs have also permeated at a local level. A study from 2007 found that “an overwhelming majority (99%) of the respondents rated environmental conservation as very important,” and that “there is overwhelming support for the conservation efforts of the government.” (Rinzin et al. 2007, pg.15) While the study uses this information to conclude that the policy made by the Bhutanese government and the work done by NGOs “appears to be just right,” a better conclusion is that Bhutanese culture is saturated with environmental awareness because of its roots in Buddhism.

While a solid portion of Bhutan’s environmental awareness can be attributed to Buddhism, it also stems from the large threat that climate change poses to Bhutan. As an agrarian country, about 57% of people living in Bhutan depend on agriculture (Chhogyel and Kumar 2018, pg.1). The effects of climate change could greatly threaten agricultural production, which is already made challenging due to the country’s mountainous topography, and therefore food security. Bhutanese farmers are generally small holders who practice sustainable integrated and subsistence farming, and are limited to the river valleys of the area. The largest natural hazard facing Bhutan is glacial lake outburst floods, or GLOFs. Glacial lakes are a very common feature at altitudes of 4,500 to 5,500 in many river basins of the Himalayas (Gurung et al. 2017).

Glacial lakes are formed as glaciers retreat, which they are doing rapidly this century, and melt-water ponds form in the basins the glaciers leave behind. These melt-water ponds grow in size until they are glacial lakes, held back by a dam made of remaining stagnant ice or other debris. GLOFs are caused by this dam breaking due to erosion, build up of water pressure, avalanches, earthquakes, or other tectonic disruptions. GLOFs can cause catastrophic societal and geomorphic impacts, and at their extreme can release millions of meters of water in a few hours, or instantaneous discharges of thousands of meters per second (Kattelman 2003). Sustained glacial melt in the Himalayas has spawned more than 5,000 glacial lakes which rest upon potentially unstable dams (Veh et al. 2020). In total, there are 567 glacial lakes in Bhutan which account for 19.03% of the total number of water bodies. Of those 567 lakes, 17 are classified as potentially dangerous (Bhutan Glacial Lake Inventory 2021). Because of how the natural topography limits agriculture, GLOFs pose a massive threat to subsistence farming that remains in the valleys below these unstable dams. Climate change will make these catastrophic floods much more common, which is not only a huge economic liability but also a danger for any Bhutanese citizen.

Maintaining and protecting biodiversity and forest cover are constitutionally mandated goals. The 2008 constitution of Bhutan states that a minimum 60% of the total land in Bhutan be forested. About 50% of the land is already designated as protected areas. Bhutan's efforts to design environmentally-motivated legislation are relatively recent. The major legislations concerning the protection of the environment were all created in the past 30 years: the Forest and Nature Conservation Act of 1995, the Environmental Assessment Act in 2000, the National Environment Protection Act in 2007, the Waste PRevention and MAnagement Act in 2009, and the Water Act of 2011. The first hydropower plant in Bhutan was commissioned in 1988, and with the growth of environmental policy, the focus on hydropower has also grown.

Bhutan's development story is a unique one for South Asia - the country was never colonized. Isolated by its topography and environment, Bhutan was a challenging land to outsiders. Bhutan did suffer several invasions, China in 1720 and the British in 1772 and 1864, but had no resources worth harnessing at the time. Bhutan has a small population that was all subsistence farmers, its land was extremely mountainous, and the abundant rivers were of no use until much later with the rise of hydroelectric power plants. The low population density meant that there was no surplus labor, which is required to kickstart industrialization. Bhutan's relative lateness to industrialization has resulted in a heavy reliance on trade and foreign aid.

The first recorded trade between Bhutan and India dates back to the period of British hegemony in India. Following India's independence, the two countries signed the Treaty of Friendship and Cooperation in 1949, which was then updated in 2007. The India-Bhutan relationship has been cordial since New Delhi established diplomatic relations in 1968 (Sarki 2019). One of the motivating factors for Bhutan to build this relationship was the geopolitical tension felt from China. The annexation of Tibet by China spurned Bhutan's reevaluation of its previously traditional policy of isolation. Consequently, Bhutan developed stronger ties with India and the process of socio-economic development became backed by Indian aid. With no history of geopolitical disputes or any rivalries between the two countries, their trade policies and diplomacy have been touted as a mutually beneficial relationship.

> "Diplomacy and trade between India and Bhutan have flourished in a framework of mutual trust. Highly prioritized cooperative arrangements have taken place in the hydropower sector, hailed as the centerpiece of the countries' relations. Given the large number of direct and indirect benefits that have been reaped in the past for both countries, in many ways the hydropower sector serves as an excellent example of a mutually beneficial relationship between two partner countries." (Tortajada, Salkani, 2008, pg.117)

Tortajada and Salkani posit that Bhutan can be seen as a "model state" for small-state diplomacy because of the success they have had in cooperating on issues like energy and water, which are non-traditional. They also argue that the energy cooperation between India and Bhutan is an example of the future of geopolitical relations in South Asia, and that this cooperation could help "capitalize one of the largest energy markets in the world." South Asia is the fastest-growing region in the world and houses a quarter of the global population, making energy availability a challenge. Collaboration between countries in this region has been identified by development agencies as the most cost-effective way to address the growing demand for energy (Tortajada and Molden 2021, pg.362). The arguments of Tortajada, Salkani, and Molden are popular ones, but the veil of mutualism begins to fall apart once hydropower agreements between India and Bhutan are closely examined.

Case Study - The Mangdechhu Hydroelectric Project:

The Mangdechhu hydroelectric power (HEP) project is a run-of-river power plant, built on the Mangde Chhu River in the central Trongsa Dzongkag district of Bhutan. The Indian government funded the project through a 70% loan and 30% grant with an interest rate of 10% and a 12-year repayment period (Vasudha Foundation 2016). In 2016 at the time of the agreement the revised cost estimate was Rs40.20bn or $602.7m USD. The Mangdechhu HEP is one of ten hydroelectric projects developed as a part of the Royal Government of Bhutan's initiative to create 10GW of hydropower by 2020. The project was commissioned in 2017 and completed in 2019. The underground excavation necessary for constructing the Mangdechhu HEPP ran into geological challenges, as any excavation will in the Himalayas.

> "Underground excavation in the Himalayas is a tough challenge attributed to complexity of the geology brought by the complex orogenic processes. The underground powerhouse caverns of the Mangdechhu hydroelectric project, were housed within quartzite of high strength and stability. However, encountering a shear zone of 1.5 to 2 m thickness with 2 to 6 m thick associated fracture zone, formation of 10.5 m high cavity and sudden increase in deformations and stresses, were the major geological and geotechnical challenges faced during the excavation of caverns." (Mishra et al. 2018, p.39)

Despite the geological challenges involved and the land taken from both citizens and forest, the Mangdechhu HEP has been celebrated as a success. The project took Bhutan's hydropower capacity to 2,326MW which is an increase of 44.8%, and was awarded the Institution of Civil Engineers' Brunel Medal for 2020 (Murray 2021).

Analysis of Hydropower and India's Role

While the Mangdechhu HEP is touted as a success story and the relationship between India and Bhutan is praised, there are certain valid concerns among a section of Bhutan over India's role in such HEP agreements. One of the complaints Bhutan has is that India buys cheap electricity from hydroelectric projects. An example of this is in 2017 when the tariff rate on imported electricity from the Tala hydroelectric project by India was much below the domestic market price in India. Another concern is unfairness in the guidelines issued by the Indian Cross Border Trade of Electricity (CBTE). Bhutan's Sustainable Hydropower Development Policy of 2008 called for a minimum shareholding of 51% by the Royal Government of Bhutan in any projects involving India. <<**find figure for stake that India has in Mangdechhu or other plants**>>

There are also concerns over the delay of completion of HEP projects. While delays in HEP projects are mostly due to geological challenges such as those encountered in the building of the Mangdechhu plant, they have a direct impact on domestic revenue. For example, loans granted by India to Bhutan increase by 10% which add to the growing debt of the country (Gross National Happiness Commission 2020). Dasho Karma Ura, President of the Center of Bhutan Studies and Gross National Happiness Research stated, "It is important that hydropower, which is [a] key issue for the Bhutanese people also be looked at more quickly [...] public opinion in Bhutan was beginning to question the viability of the debt incurred by the projects."

Critical studies of Indo-Bhutanese energy agreements are hard to find. One such study was performed by the Vasudha Foundation in 2016, a New Delhi-based NGO, titled "A Study of India-Bhutan Energy Cooperation Agreements and The Implementation of Hydropower Projects in Bhutan." According to their report, the primary challenge faced by Bhutan is hydropower debt. This was an accurate assessment. As of now three HEP projects - Mangdechhu, Punatsangchhu-I and II - have put Bhutan in debt by 12,300 crore (US $162.9 billion), which accounts for 77.52% of the country's total debt. The issue of hydropower debt might seem contradictory. The hydropower sector has benefited Bhutan's economy to a great extent through substantial contributions to the national revenue. Before the economic downturn from the COVID-19 virus, Bhutan's GDP was growing at a rapid rate. NGOs like the World Bank list Bhutan as a "development success story" because of this economic growth. However, the large initial investment costs of these massive HEP projects has put Bhutan in a green-energy debt trap, and India isn't helping.

> Close to 85% of Bhutan's trade is with India consisting mainly of HEP. On an average Bhutan has been supplying around 5,000- 5,500 million units every year to India. India also exports a limited unit to Bhutan, mainly during the dry season in Bhutan. In 2017, Bhutan imported electricity worth 74.94 million NU (US $0.99 million) from India and exported electricity of worth 11,983.49 million NU (US $1585 million). This made trade balance in its favor in around 11,908.55 million NU (US $1575 million). However, when other traded goods are included, the balance tilts in favor of India. (Ranjan 2020)

Bhutan's trade deficit stems from this imbalance in export and import, which is calculated and maintained by India.

> Much of the construction supplies for hydropower projects are purchased in India by Indian contractors and imported to Bhutan. Even when goods such as cement, transport trucks, food supply, vegetables and fruits are available in Bhutan, contractors of various projects have been found to import them from India. (Vasudha 2016)

With the increase in hydropower initiative taken by private companies from India, the Bhutanese feel that they are not equal participants in the exploration of water resources of their own country. According to the Vasudha study, the Bhutanese private sector has not reaped the benefits of hydropower development. With this comes the concern over the emergence of a secondary Indian economy in Bhutan. On top of the influx of private Indian energy companies, the HEP projects have not employed enough local Bhutanese citizens.

> According to a release by the Indian Embassy in Bhutan in 2015, "There are about 60,000 Indian nationals living in Bhutan, employed mostly in the hydro-electric power and construction industry. In addition, between 8,000 and 10,000 daily workers enter and exit Bhutan every day in [the] border towns." The rate of unemployment in Bhutan was 2.1% in 2013 and 2.9% in 2014. At many project sites and after the start of operations, some of the locals allege that they have not been provided with the necessary training to take up jobs in the sector. This is being contradicted by the Royal Monetary Authority Report of 2016-17 which says that the hydropower sector has contributed to generating direct and indirect employment opportunities for the Bhutanese. (Ranjan 2020)

The Vasudha study reported that although joint HEP projects were implemented with assistance by representatives from both governments, their analysis revealed that control of management is skewed in the favor of India. A disproportionate percentage of decision-making roles within these projects are held by Indian citizens, and "[f]urthermore, the planning, designing and management of projects, implemented under the India-Bhutan energy cooperation agreement, and all major construction and supply contracts are handled by Indian agencies." (Vasudha 2016, pg.ix)

Another issue noted by the Vasudha study was access to information. Projects implemented with assistance from India have restricted access to information relating to agreements and hydropower projects, to the point where there is minimal access to essentially basic information. Projects implemented with assistance from other governments and NGOs, like the Asian Development Bank (ADB) have "proactively" disclosed this basic information. The Environmental Impact Assessment (EIA) conducted for Punatsangchhu I, Punatsangchhu II, and Mangdechhu are all not publicly available. Yeshey Dorji, an environmentalist and wildlife photographer in Bhutan claims that the big problem is that in Bhutan hydropower projects are implemented sectryle, some even without environmental clearances. The reason for this secrecy is the involvement of Indian companies, and the Bhutanese officials fear of upsetting India by creating any problems over construction of HEPs.

The environmental concerns about the hydropower-heavy relationship between Bhutan and India are extremely pressing, and can't often be found anywhere in the state's environmental rhetoric. The Former Economic Affairs Minister of Bhutan, Norbu Wangchuk, stated that as long as all hydropower projects are run-of-river, they are not harmful to the environment. Norbu claimed that the social and environmental costs of HEP are low, and that the construction of such projects only involves temporary harm. While it is true that run-of-river hydropower projects are less invasive than dam or reservoir projects, the "temporary" harm they create can be massive, especially in terms of conservation. For example, the project site for two Punatsangchhu hydroelectric projects were part of the habitat of the critically endangered White Bellied Heron. Current population estimates for the species are between 50 to 200 remaining birds. (Birdlife.org)

HEPs in Bhutan are extremely dangerous constructions because of how tectonically active the Himalayas are. Bhutan lies on the Himalayan fault line which is responsible for the earthquake that destroyed Nepal in 2015. Because they hold huge ice reserves and massive glacial lates, the Punatsangchhu and Mangdechhu river basins possess the highest risk GLOFs in Bhutan (V, pg.36) Unfavorable monsoon conditions also pose the threat of landslides, which are amplified during project construction due to loose soil structure. Removal of forest cover, which is necessary to build HEP projects, further loosens soil structure and increases the possibility of erosion. The total land required to build the Mangdechhu HEP was 803 acres, 733 of which were Government Reserve Forests, and 70 of which were private lands belonging to 49 households. Ten families lost both home-steads and agricultural lands. (Vasudha 2016)

Perhaps the most disturbing impact of India-lead hydroelectric projects outlined in the Vasudha study is the removal of land and resources from the local people of Bhutan. Focus group discussions were held with community members from villages affected by HEP projects. These community representatives stated that government authorities held a meeting with the affected villages to inform them of the upcoming hydropower project but did not seek consent for their lands or implementing the project in their neighborhood. Most of the meeting discussed the benefits of electrification and infrastructure the projects would bring. The district and project authorities did not discuss any potential adverse impacts of hydropower plants during these consultative meetings. The planned, and now paused, 540 MW Amochhu Reservoir Hydroelectric Project would likely displace the oldest indigeonous community of Lhop or Doya people from their home.

Private land can be acquired for developmental projects under the Land Act of 2007, which requires that when private land is acquired, landholders are given a choice between replacement land and monetary compensation (pg.38). Resettlement plans were made by the state and did not consult the families who were displaced. Replacement lands are predominantly on hilltops that are scarce in water and less fertile. Many villagers interviewed in the Vasudha study mentioned that their families had to work the new barren lands for years before receiving healthy crop yields. Families with small holdings who owned under 10 decimals of land were not protected under the Land Act of 2007 and weren't eligible for replacement land, meaning they lost all the land they owned. When taking into consideration the large proportion of subsistence farmers in Bhutan, land ownership is analogous to food security.[2]

The Sankosh HEP project is a reservoir-based dam planned to be built by **<year>**. Sankosh has a potential of 2,500 MW which is twice as much as any of Bhutan's current HEP projects. Part of the Sankosh proposal calls for a canal that would run through the Indian state of West Bengal into the Teesta River. According to the Wildlife Protection Agency of India, the 60-meter-wide and 141-kilometer-long canal would cut through the Buxa tiger reserve. There is no information in the public domain on whether or not this canal still remains necessary. Once

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[2] The forced removal of people from their land is not a new concept for Bhutan. While Jigme Singye Wangchuck shaped the constitutional monarchy in Bhutan with prompting the drafting of a democratic constitution and enacting forced royal abdication, his reign also oversaw the expulsion of thousands of Lhotshampa refugees (Hutt 2005).

completed, the Sankosh project will be the most expensive hydropower project ever undertaken in Bhutan at a total of BTN 115 billion (US $15.2 billion).

The consequences of the Sankosh plant could be huge. Reservoir hydropower is considerably more invasive than run-of-river plants. A dam and reservoir can alter natural water temperatures, water chemistry, river flow characteristics, and silt load, all of which could have negative effects on the biodiversity of the area. The construction necessary could be seismically large enough to trigger GLOFs in an area that's already tectonically sensitive. A project of such magnitude will require large amounts of land, and if the state follows their previous actions, this land will be taken from forest cover and local farms. In the media, the Sankosh reservoir is being praised for its size, which is said will help Bhutan's economy and lead to job growth.

Discussion of hydropolitics & conclusion

Hydropower presents an opportunity for countries like Bhutan to receive foreign aid and further their own economic development. But with international hydropower trade comes hydropolitics and the chance for cross-border power asymmetries to grow, such as in the case of India and Bhutan. Hydropower development as a green growth strategy and the rise in dam construction in developing countries is driven by neoliberal ideologies that encourage depoliticized development policy. This depoliticization frames hydropower as a neutral source of energy, and ignores the socio-economic and environmental detriment of large hydroelectric projects. Consentual politics and the depoliticization of hydropower present HEP projects as a logical, clean energy source. Huber and Joshi (2015) argue that "[t]he dominant discourse legitimizes large hydro as clean, reliable and affordable and positions dam development as the only 'moral alternative to fossil fuel-based electricity.'" In reality, there is nothing neutral about dam building, especially in Bhutan.

Bhutan is in a delicate balance: the global hydropower market has set them in a position where they could make huge financial and developmental gains. But India's control over the hydropower sector of Bhutan sets the country at a large economic disadvantage. Furthermore, hydropower development is in almost complete contradiction with the state's guiding values of environmental development. For a government claiming to have value-driven and not economically-driven development policy, the implications and risks involved in building the Punatsangchhu, Mangdechhu, Sankosh, and all current and future HEP projects go against the values Bhutan's leaders claim. Bhutan is more than the small, picturesque, environmentally-conscious country the media portrays it to be: it is a country that will set the stage for the future of environmentally-conscious development.

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